支援SnapDrive UNIX的多重路徑存取

Snapdrive for Unix

NetApp
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支援SnapDrive UNIX的多重路徑存取

適用於UNIX的支援FC多重路徑存取、使用標準的多重路徑軟體解決方案存取儲存系統。SnapDrive透過多重路徑、您可以在主機與儲存系統之間設定多個網路路徑。

平台	多重路徑解決方案
AIX	適用於FC • NativeMPIO • DMP

如果有一條路徑故障、FC流量會繼續保留在其餘路徑上。如果主機有多個LUN路徑、則需要使用多重路徑、並讓基礎路徑對使用者透明。如果SnapDrive 設定並支援針對UNIX作業使用的組態檔中指定的多重路徑解決方案、SnapDrive 則適用於UNIX的支援將使用指定的多重路徑解決方案。

啟用多重路徑

若要在SnapDrive UNIX中使用多重路徑、您必須完成安裝和啟動HBA的步驟。如果您使用的是非NetApp多重路徑解決方案、則必須從HBA廠商的網站下載軟體與應用程式套件。

您的系統必須執行目前版本的FC和iSCSI系統元件。

"NetApp 互通性對照表工具"

步驟

1. 安裝適當的主機公用程式軟體之前、請先安裝支援的HBA。

   適用於UNIX的支援多廠商HBA。SnapDrive

2. 啟動HBA服務。

   如果HBA服務未執行、當您執行SnapDrive 支援UNIX的指令時、會顯示下列錯誤訊息、例如「SnapDrive 不支援儲存的建立」和SnapDrive 「不支援的組態準備LUN」：

   ```
   0001-876 Admin error: HBA assistant not found
   ```

   您必須驗證所需的路徑數目是否已啟動並正在執行。您可以使用主機公用程式軟體隨附的sanlun公用程式來驗證路徑。例如、在FC多重路徑組態中、您可以使用「暫停FCP show介面卡-v」命令。

   在下列範例中、有兩個HBA連接埠（fcd0和fcd1）連接至運作中的主機（連接埠狀態）。您只能擁有一個HBA或iSCSI啟動器、而且必須提供多個路徑來設定多重路徑、才能通往目標LUN。

```
# sanlun fcp show adapter -v
adapter name: fcd0
WWPN: 50060b000038c428
WWNN: 50060b000038c429
driver name: fcd
model: A6826A
model description: Fibre Channel Mass Storage Adapter
(PCI/PCI-X)
serial number: Not Available
hardware version: 3
driver version: @(#) libfcd.a HP Fibre Channel
ISP 23xx & 24xx Driver B.11.23.04
/ux/core/isu/FCD/kern/src/common/wsio/fcd_init.c:Oct 18
2005,08:19:50
firmware version: 3.3.18
Number of ports: 1 of 2
port type: Fabric
port state: Operational
supported speed: 2 GBit/sec
negotiated speed: 2 GBit/sec
OS device name: /dev/fcd0
adapter name: fcd1
WWPN: 50060b000038c42a
WWNN: 50060b000038c42b
driver name: fcd
model: A6826A
model description: Fibre Channel Mass Storage Adapter
(PCI/PCI-X)
serial number: Not Available
hardware version: 3
driver version: @(#) libfcd.a HP Fibre Channel
ISP 23xx & 24xx Driver B.11.23.04
/ux/core/isu/FCD/kern/src/common/wsio/fcd_init.c:Oct 18
2005,08:19:50
firmware version: 3.3.18
Number of ports: 2 of 2 port type: Fabric
port state: Operational
supported speed: 2 GBit/sec
negotiated speed: 2 GBit/sec
OS device name: /dev/fcd1
```

如果在主機上啟用多重路徑、則相同LUN的多個路徑會顯示出來。您可以使用「左LUN show all」命令來驗證路徑。在下列範例中、您可以找到多個路徑、指向同一個LUN（「fish：/vol/vol1/LUN」）：

```
filer:          lun-pathname        device filename  adapter    protocol
lun size              lun state
    filer_1:        fish/vol/vol1/l1      hdisk36          fcs0
FCP       5m   (5242880)        GOOD
    filer_2:        fish/vol/vol0/lunKr   hdisk39          fcs0
FCP       100m (104857600)      GOOD
    filer_3:        fish/vol/vol0/testaix hdisk2           fcs0
FCP       200m (209715200)      GOOD
    filer_4:        fish/vol/vol1/l1      hdisk45          fcs1
FCP       5m   (5242880)        GOOD
```

3. 在FC組態中、使用交換器分區組態來分區主機HBA連接埠和目標連接埠。
4. 安裝並設定適當的FC或iSCSI交換器。
5. 檢查SnapDrive UNIX堆疊需求的功能。
6. 安裝或升級SnapDrive 適用於UNIX的功能。
7. 驗SnapDrive 證UNIX版的安裝過程。
8. 找到「snapdrive.conf」檔案路徑。
9. 在「snapdrive.conf（檔案）”中設定下列組態變數：
   - 多重路徑類型_
   - 缺省傳輸
   - fstype_'
   - vmtype_'

     對於每個主機、多重路徑類型、傳輸類型、檔案系統和Volume Manager類型彼此相依。下表說明所有可能的組合：

主機平台	預設傳輸類型	多重路徑類型	**FSType**	**vmtype**
AIX	FCP	絕對	JFS2或jfs	LVM
nativempio	JFS2或jfs	LVM	DMP	VxFS

+ 上表提供了所支援的「多重路徑類型」、「預設傳輸」、「*fstype*」及「*vmtype*」組態變數值。

10. 儲存「snapdrive.conf」檔案。

    UNIX版的還原功能會在每次啟動時自動檢查此檔案。SnapDrive您必須重新啟動SnapDrive UNIX版的功能、變更才會生效。

相關資訊

"NetApp支援"

"NetApp互通性"

"《AIX Host Utilities 6.0安裝與設定指南》"

重新整理DMP路徑的理由

在具有FC和DMP組態的主機上、「SnapDrive 停止儲存刪除- LUN」命令可能會掛起。

由於下列元件的安裝或組態不當、命令可能會當機：

- NTAPasl
- Veritas堆疊（儲存基礎）
- 多重路徑授權

啟用、停用或新增任何FC路徑之後、您必須正確重新整理DMP路徑資訊。若要重新整理DMP路徑、請依照下列順序執行下列命令。

平台	命令
AIX	• 《cfgmg • "vxdisk scandon"

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